Linear algebra section 7

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Problem 1. *Find all* 3×3 *matrices such that the geometric multiplicity of* 5 *is 1, and the algebraic multiplicity of* 5 *is 3. Can such a matrix be symmetric?* ***Hint: Try findinding dimension of kernel of*** $A - \lambda I$.

Note: The above can be generalized to $n \times n$, if we want to make the problem more difficult.

Problem 2. *Give an example of* 2×2 *matrix with integral non-zero entries whose eigenvalues are* $5, 7$.

Problem 3. *Let* $L : \mathbb{R}^n \to \mathbb{R}^n$ *be an invertible matrix such that for a vector* v, $v, Lv, \ldots, L^{n-1}v$ *are linearly independent. Let* $f(x) = a_0 + a_1 x + \ldots + a_n x^n$ *be the characteristic polynomial of* L, *then* $a_0 \neq 0$, $a_1, \ldots a_{n-2} = 0$.

Note: maybe we can make it a 4×4 matrix.